

Arushi Sagar

Ghaziabad, Uttar Pradesh, India | ✉ arushisagar18@gmail.com | ☎ +91-7488358904

🐙 GitHub | 🔗 LinkedIn

Education

KIET Deemed to be University

B.Tech in Computer Science & Engineering (AI & ML)

Expected Graduation: 2028

CGPA: 8.27

Technical Skills

Programming: Python, C++, SQL

Machine Learning: Supervised Learning, Feature Engineering, Model Evaluation, XGBoost, SHAP

Deep Learning & GenAI: PyTorch, Transformers, NLP, LangChain, LLM Applications, Prompt Engineering, RAG

Data & Visualization: Pandas, NumPy, Plotly, Streamlit, Folium

Backend & APIs: FastAPI, REST APIs, Pydantic, JSON, HTTP

Databases: MySQL, SQLite

Tools & Cloud: Git, GitHub, Linux, VS Code, Google Colab, Power BI, AWS

Projects

PipelineGuardian – ML Experimentation Auditor

Jul 2026

GitHub | Live Demo

- Built an auditor for ML notebooks and scripts detecting data leakage, reproducibility gaps, validation mismatches, and train-test split errors using AST-based static analysis.
- Combined deterministic checks with LangChain-powered LLM reasoning and deployed as a CLI tool, FastAPI backend, and Vercel web application with test coverage.

Tech: LangChain, FastAPI, Python AST, Pydantic, Groq, Pytest, Vercel

Transformer from Scratch – Neural Machine Translation

Jun 2026

GitHub

- Implemented the original Transformer architecture in PyTorch from scratch for English-German neural machine translation, including multi-head attention, positional encodings, and beam search decoding.
- Achieved 35.25 BLEU on Multi30k and conducted a 12-configuration ablation study on attention heads, depth, and FFN dimensions with attention visualization tooling.

Tech: PyTorch, NumPy, TorchText, Multi30k, SacreBLEU

InfraUrban – Urban Heat Intelligence Platform

Jan 2026 – Mar 2026

GitHub | Live Demo

- Trained an XGBoost heat-risk model across 24 zones in 6 Indian cities using forward time validation, improving RMSE by 61.9% over a lag-1 baseline.
- Built SHAP explainability, warming trend analysis, and deployed results through a Streamlit dashboard with geospatial visualization.

Tech: XGBoost, SHAP, SQLite, Streamlit, Folium, Plotly

Problem Solving & Open Source

- Solved 100+ DSA problems on LeetCode spanning arrays, hashing, trees, and graph concepts.
- Contributed to Microsoft FLAML (4.4k+ stars) and render-engine, including a feature PR for render-engine that added include/exclude glob filtering with test coverage, merged after four rounds of maintainer review.

Certifications

AWS Certified Data Engineer – Associate | AWS Certified Machine Learning – Associate | AWS Certified Cloud Practitioner – Foundational